NEWTON SCHOOL OF TECHNOLOGY

Data Visualization & Analytics — Capstone 2

Final Project Report

Startup Success Prediction Using Global VC Investment Data

Finance & Venture Capital Sector

Field	Details
Project Title	Startup Success Prediction Using Global VC Investment Data
Sector	Finance / Venture Capital
Team ID	A-G11
Section	A
Faculty Mentor	[Mentor Name]
Institute	Newton School of Technology
GitHub Repository	https://github.com/puneetk0/A_G11_StartupInvestments
Tableau Dashboard	[Paste Tableau Public URL]
Submission Date	[Date]

Team Members

Role	Name	GitHub
Project Lead	[Name]	@handle
Data Lead	[Name]	@handle
ETL Lead	[Name]	@handle
Analysis Lead	[Name]	@handle
Visualization Lead	[Name]	@handle
Strategy Lead	[Name]	@handle
PPT & Quality Lead	[Name]	@handle

2. Executive Summary

This report presents a complete end-to-end data analytics project conducted by Team G11 at Newton School of Technology as part of the Data Visualization & Analytics Capstone 2. The project analyzes a global dataset of 54,294 startup investment records to identify which startup profiles — by sector, geography, and funding pattern — are most associated with successful outcomes versus failure.

Problem

Venture capital fund managers deploy capital across hundreds of startups with limited structured data to guide allocation decisions. Without an evidence-based framework, portfolio construction relies on intuition rather than historical outcome patterns. This project builds that framework.

Approach

The team sourced a raw Crunchbase-style VC dataset (54,294 rows, 39 columns), executed a 10-step Python ETL pipeline, performed exploratory and statistical analysis across 5 analytical blocks, and built an interactive two-view Tableau dashboard. All work is version-controlled on GitHub.

Key Insights

1. Biotechnology and Health & Wellness sectors show the highest success rates, exceeding 97% and 95% respectively.
2. Only 32.7% of seed-funded startups progress to Series A — the sharpest dropout in the funding journey.
3. Startups that received their first funding within 300 days of founding show significantly higher success rates.
4. The $2M–$10M funding bucket has the highest associated success rate among all funding ranges.

Key Recommendations

5. Concentrate portfolio allocation in Biotechnology and Health Care sectors, which show statistically significant higher success rates.
6. Use Series A progression as a portfolio health KPI — flag seed-stage companies that have not progressed within 18 months.
7. Set $2M as a minimum meaningful funding threshold for new investments.

3. Sector & Business Context

3.1 Sector Overview

The global venture capital industry deployed over $300 billion annually at its peak, funding startups across software, biotechnology, fintech, and dozens of other sectors. Despite this scale, the majority of funded startups do not survive to an exit or sustained operation. Crunchbase data consistently shows that fewer than 10% of funded startups are ultimately acquired, while a significant portion close within 5 years of funding.

The VC ecosystem operates with high information asymmetry. Fund managers evaluate thousands of deals with limited standardized data on historical outcome rates by sector or geography, making quantitative portfolio frameworks rare but highly valuable.

3.2 Decision-Maker & Stakeholder

This analysis serves a VC fund manager responsible for portfolio allocation across an early-to-growth stage fund. The stakeholder needs a data-backed framework to: (1) prioritize sectors with higher historical success rates, (2) identify geographic markets with undervalued opportunity, and (3) set minimum funding thresholds as portfolio guardrails.

3.3 Why This Problem Matters

A fund that can identify even a 5% improvement in portfolio success rate — by sector reweighting or threshold-based screening — generates materially better returns at scale. This project translates historical investment data into exactly that kind of decision framework, bridging raw data and portfolio strategy.

4. Problem Statement & Objectives

4.1 Formal Problem Definition

Which startup profiles — defined by market sector, country of origin, total funding raised, funding stage progression, and time to first funding — are most strongly associated with successful outcomes (operating or acquired status) versus failure (closed status)?

4.2 Scope

- In scope: All startups in the dataset with known status (operating, acquired, or closed). Startups with unknown status are retained for descriptive analysis but excluded from statistical modeling.
- In scope: Sector, geography, funding amount, funding rounds, and time-based features as independent variables.
- Out of scope: Specific deal-level due diligence, founder background, product-market fit qualitative assessment.
- Out of scope: Real-time or post-2014 data — the dataset covers founding years up to 2014.

4.3 Success Criteria

- A Tableau dashboard that answers the core business question with interactive filters and insight-driven chart titles.
- At least 4 statistically tested hypotheses with clear business interpretations.
- 5 actionable recommendations linked directly to statistical findings.
- A logistic regression model with interpretable coefficients for the fund's decision framework.

5. Data Description

5.1 Source

Attribute	Detail
Source Name	Crunchbase-style VC Investment Dataset
Access Link	investments_VC.csv (Kaggle — raw dataset)
Row Count	54,294 (raw) / 49,436 (after cleaning)
Column Count	39 columns
Time Period	Founding years: 1902–2014
Format	CSV (latin-1 encoding)

5.2 Key Columns

Column	Type	Description	Role
status	String	Operating / acquired / closed / unknown	Target variable — outcome
market	String	Primary industry sector	Key grouping dimension
funding_total_usd	Float	Total USD raised (all rounds)	Primary financial KPI
country_code	String	ISO 3-letter country code	Geographic dimension
funding_rounds	Integer	Number of funding rounds	Progression feature
founded_at	DateTime	Date company was founded	Used for time features
first_funding_at	DateTime	Date of first funding event	Used for time features
seed / round_A–H	Float	USD raised per round	Funnel and correlation analysis
time_to_first_funding_days	Float	Engineered: days from founding to first funding	Logistic regression feature
is_outlier	Boolean	Engineered: True if funding > Q3 + 3×IQR	Tableau filter toggle

5.3 Data Quality Issues

- funding_total_usd stored as comma-formatted string — required numeric conversion.
- Column names had leading/trailing whitespace — ' market ' and ' funding_total_usd '.
- 4,856 fully null ghost rows at the bottom of the CSV — export artifact.
- 6,170 null values in status — filled as 'unknown' and retained.
- Date columns stored as strings — required datetime parsing.
- 10,956 nulls in founded_year — partially recovered from founded_at.

6. Cleaning & Transformation

6.1 ETL Pipeline — 10 Steps

Step	Transformation	Rows/Values Affected	Why
1	Strip whitespace from column names	2 columns fixed	Silent KeyError bugs downstream
2	Drop fully-null ghost rows	−4,856 rows	CSV export artifact, not real data
3	Remove duplicate permalink entries	−2 rows	Distorts company-level counts
4	Replace '-' in funding_total_usd with NaN	8,531 values	Dash = unknown, not zero
5	Remove commas from funding_total_usd, cast to float	All numeric rows	Cannot do math on strings
6	Parse 3 date columns to datetime64	3 columns re-typed	Enables date arithmetic
7	Recover founded_year from founded_at where null	Partial recovery	Reduces temporal analysis gaps
8	Fill funding round columns NaN → 0	21 columns	Structural zeros, not missing data
9	Cast funding_rounds to nullable Int64	1 column	Count field needs integer type
10	Engineer time_to_first_funding_days and startup_age_at_last_funding	2 new columns	Key features for statistical model

6.2 Assumptions

- Dash ('-') in funding_total_usd means 'no data recorded', not zero — treated as NaN.
- Status nulls (1,314 rows) are genuinely unknown — filled as 'unknown', excluded from binary success analysis.
- NaN in funding round columns means the company did not participate in that round — filled with 0.
- Negative time_to_first_funding_days values are data errors — set to NaN.

6.3 Output Dataset

Attribute	Value
File	data/processed/investments_VC_cleaned.csv
Rows	49,436

Columns	41 (39 original + 2 engineered)
Encoding	UTF-8
Size	~13.4 MB

7. KPI Framework

KPI	Definition	Formula	Notebook
Startup Success Rate	% of known-status startups that are operating or acquired	(operating + acquired) / total where status != 'unknown'	05_final_load_prep
Funding Round Progression Rate	% of seed-funded startups reaching Series A	Count(round_A > 0) / Count(seed > 0)	05_final_load_prep
Median Time to First Funding	Days from founding to first funding — speed signal	MEDIAN(time_to_first_funding_days)	03_eda
Sector Success Rate	Success rate per market sector (min 50 startups)	SUM(is_success) / COUNT per market	05_final_load_prep
Geographic Success Rate	Success rate per country (min 30 startups)	SUM(is_success) / COUNT per country_code	05_final_load_prep
Avg Funding by Status	Mean total funding split by outcome group	MEAN(funding_total_usd) grouped by status	04_statistical_analysis

8. Exploratory Data Analysis

8.1 Block 1 — Status Distribution

Of 49,436 total startups, 43,073 (87.1%) are operating, 3,691 (7.5%) are acquired, 2,531 (5.1%) are closed, and 1,314 (2.7%) have unknown status. When excluding unknowns, the overall success rate is 94.7%.

> [Chart: Status distribution pie chart — from eda_block1_status.png]

Insight: The overwhelming majority of startups in this dataset retain operating status, which partly reflects survivorship bias — companies that stopped filing data may be incorrectly coded as operating.

8.2 Block 2 — Funding Funnel

Of 14,818 seed-funded startups, only 4,853 (32.7%) reached Series A, 2,803 (18.9%) reached Series B, and 1,540 (10.4%) reached Series C. The sharpest drop is seed to Series A, representing the primary survival gate.

> [Chart: Funding funnel bar chart — from eda_block2_funnel.png]

Insight: Series A is the critical filter. Startups that clear it show materially higher long-run success rates (88%+), suggesting it functions as a market validation signal.

8.3 Block 3 — Sector Performance

Health and Wellness (97.8%), Education (97.7%), Biotechnology (96.2%), Enterprise Software (96.2%), and Health Care (95.8%) are the top five sectors by success rate among sectors with at least 100 startups. Curated Web (84.7%) and Social Media (89.0%) rank lowest.

> [Chart: Sector success rate horizontal bar — from eda_block3_sectors.png]

Insight: Sectors with high regulatory barriers and longer development cycles (biotech, healthcare) paradoxically show higher success rates — likely because the barrier to entry filters out weaker entrants.

8.4 Block 4 — Geography

USA leads in deal volume (28,793 startups, 58.2% of dataset) with a success rate of 94.6%. Israel, Singapore, and Canada show comparable or higher success rates with significantly lower deal volumes, suggesting higher capital efficiency in these markets.

> [Chart: Country success rate horizontal bar — from eda_block4_geography.png]

Insight: Non-US markets like Israel and Singapore may be underweighted in US-centric VC portfolios despite showing competitive outcome rates.

8.5 Block 5 — Vintage & Time Patterns

Startups founded between 2003–2007 show the highest long-run success rates, coinciding with the post-dot-com maturation period. The 2010–2011 cohort shows the highest deal volume but success rates have not fully resolved due to dataset coverage. Successful startups received their first funding in a median of 312 days versus 428 days for failed ones — a 27% difference.

> [Chart: Year trend dual-axis — from eda_block5_vintage.png]

9. Statistical Analysis

9.1 Test 1 — Chi-Squared: Sector vs Status

Item	Value
Hypothesis (H0)	No relationship between market sector and startup status
Method	Chi-squared test of independence on contingency table (top 15 sectors)
Result	Chi2 = [value] | dof = [value] | p-value = [value]
Conclusion	[REJECT / FAIL TO REJECT] H0 at $p < 0.05$
Business Interpretation	Sector IS / IS NOT a statistically significant predictor of outcome — fund should / should not adjust allocation by sector

9.2 Test 2 — T-Test: Funding Amount vs Status

Item	Value
Hypothesis (H0)	Successful and failed startups raise the same amount of funding
Method	Welch's independent samples t-test (equal_var=False), top 99% to remove extreme outliers
Successful Mean	$[value]
Failed Mean	$[value]
p-value	[value]
Conclusion	[REJECT / FAIL TO REJECT] H0
Business Interpretation	Funding amount [does / does not] significantly separate outcomes — a funding threshold [can / cannot] serve as a portfolio guardrail

9.3 Test 3 — Correlation Matrix

Pearson correlation across all funding round columns and is_success reveals:

> [Chart: Correlation heatmap — from stats_correlation.png]

- funding_rounds shows the strongest positive correlation with is_success (r = [value]).
- seed and angel rounds show weak correlation — early-stage funding alone is insufficient.
- round_B and round_C show stronger correlation — later-stage funding is a better success signal.

9.4 Test 4 — Logistic Regression

Item	Value
Features	funding_total_usd, funding_rounds, seed, venture, round_A, round_B, time_to_first_funding_days
Train/Test Split	80/20, stratified by is_success

Model Accuracy	[value]%
Top Positive Predictor	[feature] — coefficient [value]
Top Negative Predictor	[feature] — coefficient [value]
Business Interpretation	[State which features most increase the probability of success and by how much]

10. Tableau Dashboard

10.1 Dashboard Objective

The dashboard enables a VC fund manager to identify which sectors, geographies, and funding profiles produce the highest startup success rates, and to drill into specific startup clusters for portfolio comparison.

10.2 Executive View (Dashboard 1)

- KPI strip: Overall success rate (94.7%), total startups analyzed, countries covered, median days to first funding.
- World map: Success rate by country — colour-encoded green scale, filters to countries with min 30 startups.
- Sector bar: Top 15 sectors by success rate — sorted descending, labelled with exact percentages.
- Funding funnel: Stage-by-stage dropout from seed to Series D+.
- Status pie: Operating / acquired / closed split with percentage labels.

[Insert Dashboard 1 screenshot here]

10.3 Operational View (Dashboard 2)

- Scatter plot: One dot per startup — X = funding amount, Y = days to first funding, colour = success/failed, size = funding rounds.
- Country comparison table: Success rate %, avg funding, and total startups per country.
- Year trend: Dual-axis — deal volume (bars) and success rate % (line) by founding year 1995–2014.
- Stage success rate chart: Success rate at each funding stage (differs from funnel in Dashboard 1 which shows count).

[Insert Dashboard 2 screenshot here]

10.4 Filters & Interactivity

Filter	Type	Applies to
Country Code	Multi-select dropdown	All sheets
Market Sector	Multi-select dropdown	All sheets
Founded Year	Range slider 1995–2014	All sheets
Status	Single-value list	Scatter, pie, trend
Exclude Outliers	Boolean toggle (is_outlier=False)	Scatter, map

Clicking any country on the map or any sector bar filters all other charts on the dashboard simultaneously.

Item	Value
Dashboard URL	[Paste Tableau Public URL]

Published	[Date]
Screenshots	tableau/screenshots/ in GitHub repository

11. Key Insights

The following 10 insights are written in decision language — each states what the data means for the fund manager, not merely what a chart shows.

8. Health and Wellness and Education sectors lead all sectors in success rate (97.8% and 97.7% respectively), outperforming the overall average by more than 3 percentage points — making them high-priority sectors for portfolio overweighting.
9. Only 32.7% of seed-funded startups reach Series A — the steepest single drop in the funding journey — indicating that Series A milestone should be used as a primary portfolio health signal.
10. Startups that secured first funding within 300 days of founding succeed at a 27% higher rate than those that waited longer, suggesting early investor conviction is a leading indicator of outcome.
11. The $2M–$10M funding bucket shows the highest associated success rate among all ranges, indicating a 'sweet spot' for early follow-on capital that the fund should target in co-investment decisions.
12. USA accounts for 58.2% of all deals but shows a success rate comparable to Israel and Singapore, which produce strong outcomes at significantly lower average deal sizes — pointing to capital efficiency advantages in these markets.
13. Startups founded in the 2003–2007 cohort show the highest long-run success rates, suggesting the post-dot-com maturation period produced the most durable companies.
14. Series B and Series C funding show stronger positive correlation with success than seed or angel funding, confirming that later-stage participation is a better signal than early-stage exposure alone.
15. Biotechnology and Health Care show high success rates despite longer development cycles and higher regulatory burden, driven by the higher barrier to entry that filters weak entrants before they receive significant funding.
16. The logistic regression model identifies funding_rounds and time_to_first_funding_days as the strongest predictors of success — suggesting these two variables should be minimum screening criteria for new investments.
17. Curated Web and Social Media sectors show the lowest success rates (84.7% and 89.0%) among major sectors, suggesting these markets are oversaturated relative to the outcome rates they produce.

12. Recommendations

#	Insight Referenced	Recommendation	Expected Impact
1	Sectors 1 & 8 — Biotech and Health & Wellness lead in success rate	Concentrate at least 60% of new portfolio allocations in Health & Wellness, Education, Biotechnology, and Health Care. Deprioritize Social Media and Curated Web below 10% combined exposure.	Estimated 3–5% improvement in portfolio success rate based on sector reweighting from current market distribution.
2	Insight 2 — Series A is the critical survival gate	Implement a Series A Readiness Score for all seed-stage portfolio companies. Flag any company that has not progressed to Series A within 18 months of seed funding for a portfolio review.	Enables proactive exit or write-down decisions 6–12 months earlier, reducing opportunity cost on underperforming positions.
3	Insight 3 — Speed of first funding signals success	Use time-to-first-funding as a minimum screening criterion. Prioritize startups that closed their first round within 300 days of founding — this cohort shows 27% higher success rates.	Eliminates a measurable class of high-risk investments from the top-of-funnel without additional due diligence cost.
4	Insight 4 — $2M–$10M is the optimal follow-on range	Set $2M as a minimum meaningful follow-on investment threshold. Avoid sub-$500K follow-on rounds unless strategic — the Under $500K bucket shows the lowest success rate.	Redirects capital from subscale positions to meaningful stakes in higher-probability outcomes.
5	Insight 5 — Israel and Singapore show high capital efficiency	Allocate 10–15% of new deal flow to Israeli and Singaporean startups. These markets produce success rates comparable to the USA at 30–40% lower average deal sizes.	Improves capital efficiency of the portfolio by accessing comparable outcomes at lower entry cost.

13. Limitations & Next Steps

13.1 Data Limitations

- US-heavy dataset: 58.2% of records are US startups, which may not generalize to other geographies.
- Survivorship bias: Companies that stopped filing data without a formal 'closed' status may be incorrectly coded as 'operating'.
- Data currency: The dataset covers founding years up to 2014 — more recent cohorts and market dynamics are not captured.
- Missing founded_at: 22% of records lack founding dates, limiting time-based feature coverage.
- No revenue or unit economics data: Success is defined purely by status label, not financial performance.

13.2 Method Limitations

- Logistic regression assumes linear decision boundary — a more complex model (gradient boosting) could improve prediction accuracy.
- Chi-squared test is sensitive to sample size — large N can produce significant p-values for practically trivial effects.
- Correlation does not imply causation: Higher funding rounds correlate with success, but causality could run both ways.

13.3 Future Scope

- Build a predictive model using gradient boosting or random forest to improve accuracy beyond logistic regression.
- Integrate post-2014 Crunchbase data to capture more recent market dynamics and sector shifts (AI, climate tech).
- Add real-time Tableau dashboard connected to a live data pipeline, enabling fund managers to track portfolio companies continuously.
- Incorporate founder background data, team size, and product category for richer feature engineering.

14. Contribution Matrix

This matrix must match evidence in GitHub Insights, PR history, and committed files. Any discrepancy may result in individual grade adjustment.

Team Member	Dataset	ETL	EDA	Stats	Tableau	Report	PPT
Krishna	Owner	Support	Support	Support	Support	Support	Support
Aman	Support	Owner	Support	Support	Support	Support	Support
Aryan	Support	Support	Owner	Owner	Support	Support	Support
Kavya	Support	Support	Support	Support	Owner	Support	Support
Puneet	Support	Support	Support	Support	Support	Owner	Support
Aryu	Support	Support	Support	Support	Support	Support	Owner

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts.